

The empirical recurrence for OEIS A302080, recovered from its tail

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Abstract

OEIS A302080 records “Empirical recurrence of order 52” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 999$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 55$. Its last coefficient is $c_{52} = -63403380965376 \neq 0$, so no further tail satisfies anything shorter; any order-52 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302080 is “Number of $n \times 7$ 0..1 arrays with every element equal to 0, 1, 3 or 4 horizontally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

21, 154, 1166, 10526, 94846, 893236, 8320747, 78493707, ...

The entry states:

Empirical recurrence of order 52 (see link above)

As of the “Last modified” line on the live entry (Mar 31 2018 15:59:03, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 999$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 999$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 52: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 52.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1998$ exact terms from $n = 3$ on, it returns order 52 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{52} c_i a(n-i),$$

c_1	1	c_{14}	257249857745	c_{27}	−789911017453145	c_{40}	−620210726115284992
c_2	213	c_{15}	30544770258	c_{28}	−62729625149045873	c_{41}	48622146001426432
c_3	−68	c_{16}	−2914990801758	c_{29}	3578462765202564	c_{42}	382109012687462400
c_4	−19457	c_{17}	−256936751350	c_{30}	159820492772370973	c_{43}	−24375640944451584
c_5	287	c_{18}	26211402543449	c_{31}	−11421196681054959	c_{44}	−172818218854121472
c_6	1040284	c_{19}	1521683290983	c_{32}	−329945779151437905	c_{45}	7677835061428224
c_7	122029	c_{20}	−188993557101342	c_{33}	27140718365800704	c_{46}	54720744994308096
c_8	−37012049	c_{21}	−5647187627357	c_{34}	548691546709970504	c_{47}	−1227090568937472
c_9	−6041686	c_{22}	1100767314048605	c_{35}	−48774857041029600	c_{48}	−11252831707201536
c_{10}	939389437	c_{23}	4561819511250	c_{36}	−728613584569255200	c_{49}	21130383458304
c_{11}	155497443	c_{24}	−5203929631588993	c_{37}	66150339791078784	c_{50}	1313308573433856
c_{12}	−17756104606	c_{25}	98859909711413	c_{38}	763225806214142016	c_{51}	13209037701120
c_{13}	−2604115353	c_{26}	20021745224740908	c_{39}	−66714077919836416	c_{52}	−63403380965376

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 55$, and it is the recurrence OEIS A302080 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{52} - c_1 x^{52-1} - \dots - c_{52}$. Its constant term is $-c_{52} = 63403380965376$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 52 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 52, so $\deg q = 52$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 52, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 18 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 52, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -63403380965376 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-52 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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